

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B3. QUANTUM, ATOMIC AND MOLECULAR PHYSICS

TRINITY TERM 2016

Thursday, 16 June, 2.30 pm – 4.30 pm

10 minutes reading time

*Answer **two** questions.*

*Start the answer to each question in a **fresh book**.*

A list of physical constants and conversion factors accompanies this paper.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Do NOT turn over until told that you may do so.

1. A transition in the potassium atom ($Z = 19$) is denoted by $3d\ ^2D_{5/2} - 5p\ ^2P_{3/2}$. Explain the meaning of each part of this notation. What are the selection rules relevant to determining the allowed transitions in alkali-like atomic systems?

[7]

Several series of lines are observed in the emission spectrum of the alkali-like ion Ca^+ ($Z = 20$). There is a series of doublets, including the resonance lines at wavelengths of 393.4 nm and 396.9 nm, which terminates at 104.4 nm. A second series of doublets begins with lines at 373.7 nm and 370.6 nm and terminates at about 142 nm. Draw an energy level diagram showing the first doublets of both series, and the series limits. Label the levels with appropriate quantum numbers.

[9]

There is a third (triplet) series which terminates at about 121.8 nm. Three emission lines are observed at 849.8 nm, 854.2 nm and 866.2 nm. Explain the origins of these lines and add them to your energy level diagram labelled with appropriate quantum numbers.

[5]

The Ca^+ ion is subjected to an external magnetic field which causes the spectral lines to split into a number of separate components. For the line at 393.4 nm determine the number of components observed perpendicular to the applied field.

[4]

[The spacing of the components is not required.]

2. Show that an interaction with a Hamiltonian of the form $H = A \mathbf{J}_1 \cdot \mathbf{J}_2$ leads to an interval rule

$$E_J - E_{J-1} = AJ ,$$

where $\mathbf{J}_1$, $\mathbf{J}_2$ and $\mathbf{J}$ are angular momentum operators related by $\mathbf{J} = \mathbf{J}_1 + \mathbf{J}_2$, and E_J is an energy eigenvalue. [4]

What conditions must be fulfilled by the various interactions that determine atomic structure in order for an atom with two valence electrons to exhibit an interval rule? Give an example of a term in such an atom in which you would expect the rule to be obeyed. Draw a diagram showing the splitting of the term and label the levels with appropriate quantum numbers. [7]

The Hamiltonian for the magnetic dipole hyperfine interaction has the form

$$H_{\text{hfs}} = -g_I \mu_N \mathbf{I} \cdot \mathbf{B}_e ,$$

where g_I is the nuclear g -factor, $\mathbf{I}$ is the nuclear angular momentum operator and $\mathbf{B}_e$ is the magnetic induction at the nucleus ($r = 0$) arising from the electrons. Explain why this can be written in the form $H_{\text{hfs}} = A \mathbf{I} \cdot \mathbf{J}$. What set of good quantum numbers characterise the energy levels arising from H_{hfs} ? [5]

Measurement of the splittings between adjacent energy levels of the ground electronic configuration in the isotope ^{55}Mn gave the following 5 frequencies: 72, 145, 217.3, 289.7, 362.1 MHz. Describe how these frequencies can be interpreted and determine the values of the properties of the atomic system that can be deduced from the data.

Measurement of the splittings between adjacent energy levels of the ground electronic configuration in the isotope ^{56}Mn gave the following 5 frequencies: 85, 141, 197.4, 253.8, 310.2 MHz. Find the ratio of the g -factors of the nuclei in the two isotopes, explaining your reasoning. [9]

3. Explain what is meant by the terms *three-level laser* and *four-level laser*, giving an example of each. Discuss why there is a large difference in the threshold power which is required to achieve laser oscillation in these two classes of laser. [5]

A homogeneously broadened transition of angular frequency ω_{21} occurs between two levels which have degeneracies g_1 and g_2 . The upper level is populated at a rate R_2 and decays only by spontaneous emission to the lower level at a rate A_{21} . The lower level is populated only by this decay and has a lifetime τ_1 . The number densities for the lower and upper levels are N_1 and N_2 respectively. Write down the rate equations governing N_1 and N_2 and show that the population inversion density $N^* = N_2 - (g_2/g_1)N_1$ in the steady state is given by

$$N^* = R_2 \left(\frac{1}{A_{21}} - \frac{g_2}{g_1} \tau_1 \right). \quad [4]$$

In the presence of radiation of intensity I (at angular frequency ω), stimulated processes transfer atoms between the levels. Write down the rate equations governing N_1 and N_2 for this system, defining additional quantities that you introduce. Show that the gain coefficient at line centre has the form

$$\alpha = \frac{\alpha_0}{1 + I/I_s}.$$

Give expressions for α_0 and I_s , and state what they represent. For the case $I \gg I_s$, explain how the intensity changes as radiation propagates through the medium. [12]

A homogeneously broadened laser transition has a peak optical cross-section $\sigma(\omega_{21}) = 4 \times 10^{-15} \text{ cm}^2$, and $R_2 = 5 \times 10^{19} \text{ s}^{-1} \text{ cm}^{-3}$. The upper level has decay rate $A_{21} = 3 \times 10^6 \text{ s}^{-1}$ and $g_2 = 5$. The lower level has lifetime $\tau_1 = 2 \times 10^{-8} \text{ s}$ and $g_1 = 3$. Determine the value of α_0 .

Another transition has a comparable cross-section and its upper level is pumped at the same rate. In this case $A_{21} = 3 \times 10^6 \text{ s}^{-1}$, $g_2 = 3$, $\tau_1 = 3 \times 10^{-7} \text{ s}$ and $g_1 = 1$. What is the value of α_0 for this transition? [4]

4. In a two-state model of an atom there are two eigenstates, ψ_1 and ψ_2 , of the Hamiltonian $\hat{H}_0$ with energies E_1 and E_2 respectively ($E_2 > E_1$). Spontaneous emission is neglected. This system is subjected to a perturbation such that

$$\hat{H} = \hat{H}_0(\mathbf{r}) + \hat{V}(t),$$

where the time-dependent perturbation $\hat{V}(t) = ex\mathcal{E}\cos\omega t$ arises from an oscillating electric field of amplitude $\mathcal{E}$ along the x -direction. The general solution of this time-dependent Schrödinger equation can be written

$$\Psi(\mathbf{r}, t) = c_1(t)\psi_1(\mathbf{r})e^{-iE_1t/\hbar} + c_2(t)\psi_2(\mathbf{r})e^{-iE_2t/\hbar}.$$

Show that

$$\dot{c}_2(t) = -i\Omega \cos(\omega t)e^{i\omega_0 t} c_1(t) - \frac{iV_{22}}{\hbar} c_2(t),$$

giving the expressions for ω_0 , V_{22} , and the Rabi frequency, Ω , in terms of matrix elements of the atomic dipole (such as $\langle\psi_1|ex|\psi_2\rangle$) and other quantities. Explain why $V_{22} = 0$. [10]

The initial condition is $c_2(0) = 0$. Assuming that $c_2(t) \ll c_1(t)$ and $c_1(t) \simeq 1$, calculate the integral

$$c_2(t) = \int_0^t \dot{c}_2(t') dt'.$$

[5]

Justify the use of the *rotating wave approximation* and find an expression for $|c_2(t)|^2$. [4]

The above model applies for monochromatic radiation. Show how it can be extended to describe an atom illuminated by weak broadband radiation of spectral energy density $\rho(\omega)$ which is related to the electric field by

$$\rho(\omega) d\omega = \frac{1}{2}\epsilon_0\mathcal{E}^2.$$

Hence find an expression for the transition rate due to stimulated absorption of broadband radiation. [6]

[You may find this integral useful:

$$\int_{-\infty}^{\infty} \left(\frac{\sin \frac{1}{2}(\omega - \omega_0)t}{\frac{1}{2}(\omega - \omega_0)} \right)^2 d\omega = 2\pi t.]$$